Beyond Core GRADE. Rating risk of bias across bodies of evidence: When evidence dominated by high risk of bias studies, Considering direction of bias

When evidence is dominated by high risk of bias studies, before definitively rating down for risk of bias, GRADE users might consider the expected direction of any bias influencing results. If, in studies showing no important difference in effect, bias would have increased differences between groups, one can infer that the actual difference must be, if different at all, smaller than it appeared. Consideration of risk of bias would thus reinforce the conclusion of no difference between groups and consequently there would be no reason to rate down for risk of bias. Similarly, if effects in studies show a difference but bias would have decreased that difference, the inference would be that the true relative effect is if anything larger than that observed. There would therefore be no reason to rate down for risk of bias.

Consider for instance a meta-analysis of randomized controlled trials addressing the prevention of *Plasmodium falciparum* malaria transmission that compared the addition of primaquine to a previous regimen versus not adding the drug.[1] Results provided evidence that primaquine reduced, rather than increased, adverse events (odds ratio 0.79, 95% CI 0.63 to 1.42). Lack of blinding suggested classifying the studies at high risk of bias. However, if clinicians know that a patient is receiving an additional drug, they would be more inclined to attribute symptoms to side effects from the drug than in patients not receiving the drug. If they were blinded, there could be no such differential attribution. Thus, bias from failure to blind would have led to an overestimation of adverse events with primaquine. Considering this, the direction of bias increases our strength of inference that primaquine does not importantly increase adverse effects. Thus, review authors appropriately decided against rating down for risk of bias.

1 Yilma D, Stepniewska K, Bousema T, et al. WWARN Paediatric Primquine for P. falciparum transmission Blocking Study Group. Efficacy and Safety of Single-Dose Primaquine to Interrupt Plasmodium Falciparum Malaria Transmission in Paediatric Patients Compared to Adults: A WWARN Systematic Review and Individual Patient Data Meta-analysis. *SSRN* 2024; https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5005770